DMET 1075 - Project Document for Augmented Reality (AR) Game

- **Project Deadline:** May 17th, 2026 (Project video to be included, with the uploaded published file, all links will be in a document to be uploaded on google forms). May 18th would be your final discussions.
- **Development Platform:** Unity with ARCore
- **Grade**: 15 percent

<u>Design Requirements:</u>
Characters: 2-3 main characters with unique abilities.
Objects: Collectibles (e.g., keys, treasures) and interactive items (e.g., puzzle pieces).
Object Recognition: Use ARCore to detect and place virtual objects in the real world.
Environment Mapping: Allow players to interact with their surroundings seamlessly.

Gameplay Mechanics (4 at least to be included):

- **Exploration**: Players move in the real world to discover virtual objects.
- **Puzzle Solving**: Incorporate logic puzzles that require interaction with the environment.
- **Scoring System**: Points awarded for collecting items and completing challenges.

Workflow suggestions (to coordinate as a team):

Tech setup (Week 1) — Don't underestimate this. Getting Unity configured with ARCore and a device that supports it can take a day or two. Install the ARCore XR Plugin via Unity's Package Manager and test basic plane detection before writing any game logic. Document your design decisions early — you'll need them for the discussion on May 18.

Characters — The three suggested (Explorer, Guardian, Trickster) each map cleanly to a gameplay mechanic so they don't feel arbitrary. When building them, keep the 3D models low-poly to maintain AR performance — complex meshes tank frame rates on mobile.

The 4th mechanic — The brief lists three explicitly (exploration, puzzles, scoring). You'll need at least one more. Good options: a **time challenge** mode, a **character-switching** mechanic mid-gameplay, or a **proximity trigger** system where virtual objects react when the player physically gets close. The ability activation described in the plan above could serve as your fourth.

ARCore specifics to plan for:

- Plane detection (horizontal surfaces) is where you'll anchor most objects
- Image tracking lets you place objects on physical markers — great for the puzzle mechanic
- Light estimation makes your virtual objects look realistic in different lighting conditions

Biggest risks to watch: AR performance on lower-end Android devices, and puzzle design taking longer than expected. Aim to have a playable build by the end of Week 3 so Week 4 is purely polish and recording. Record the video in a well-lit space — AR looks much better on camera with good lighting.